A PROJECT REPORT ON

Secure Hybrid Agentic RAG for Context-Aware Banking Fraud Investigation & Decision Support

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING (Computer Engineering)

BY

Parth Pakhare	Roll No: BCOB15
Soham Sadakal	Roll No: BCOB32
Kaushal Pokarne	Roll No: BCOB33
Omkar Shelar	Roll No: BCOB64

Under The Guidance of

Dr. Jyoti Shinde

[DPU˙DYP˙Logo.png not found]

Department of Computer Engineering

DR. D. Y. PATIL INSTITUTE OF TECHNOLOGY
PIMPRI, PUNE

A.Y. 2026-2027

[UNI˙LOGO.jpg not found]

SAVITRIBAI PHULE PUNE UNIVERSITY

[DPU˙DYP˙Logo.png not found]

DR. D. Y. PATIL INSTITUTE OF TECHNOLOGY
Department of Computer Engineering

CERTIFICATE

This is to certify that the Project Entitled

Secure Hybrid Agentic RAG for Context-Aware Banking Fraud Investigation & Decision Support

Submitted by

Parth Pakhare	Roll No: BCOB15
Soham Sadakal	Roll No: BCOB32
Kaushal Pokarne	Roll No: BCOB33
Omkar Shelar	Roll No: BCOB64

is a bonafide work carried out by the students under the supervision of Dr. Jyoti Shinde and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (Computer Engineering).

Dr. Jyoti Shinde
Internal Guide
Dept. of Computer Engg.

Dr. Omkaresh Kulkarni
H.O.D
Dept. of Computer Engg.

Dr. Nitin Sherje
Principal
DIT, Pimpri

Signature of Project Co-ordinator

Signature of External Examiner

PROJECT APPROVAL SHEET

A Project Report Titled

Secure Hybrid Agentic RAG for Context-Aware Banking Fraud Investigation & Decision Support

is successfully completed by

Parth Pakhare	Roll No: BCOB15
Soham Sadakal	Roll No: BCOB32
Kaushal Pokarne	Roll No: BCOB33
Omkar Shelar	Roll No: BCOB64

at

DEPARTMENT OF COMPUTER ENGINEERING
DR. D. Y. PATIL INSTITUTE OF TECHNOLOGY
SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

ACADEMIC YEAR 2026-2027

Dr. Jyoti Shinde	Dr. Omkaresh Kulkarni
Internal Guide	H.O.D
Dept. of Computer Engg.	Dept. of Computer Engg.

Abstract

Banking fraud investigation requires an analyst to correlate transaction records, customer behaviour, device and beneficiary history, earlier cases and internal policies under tight time limits. Existing rule-based and machine-learning detectors raise alerts, but they give little explanation or context, so the analyst still collects the evidence by hand. This is slow, inconsistent and hard to audit.

This project presents a *Secure Hybrid Agentic Retrieval-Augmented Generation (RAG)* proof of concept. Incoming transactions are scored by a set of deterministic rules combined with a lightweight Isolation Forest anomaly score, and a case is opened when the combined risk crosses a threshold. When an analyst presses *Investigate*, a bounded LangGraph agent calls read-only tools and a hybrid search that merges vector similarity and keyword (BM25) ranking using Reciprocal Rank Fusion. It returns a short summary with citations to the policy passages it used, together with a typed assessment (fraud pattern, priority, evidence sufficiency, recommended action).

The central design rule is that the language model only *recommends*. The outcome (hold, escalate or close) is chosen by a deterministic, versioned rule engine, and high-impact outcomes need a human supervisor's approval. Personal data is masked before any model call, retrieved text and transaction notes are treated as untrusted, access is role-based, and every state change is written to an append-only hash-chained audit log. All data is synthetic and all actions are simulated.

The Stage 1 outcome is a working end-to-end prototype of this flow. The report describes the problem, requirements, design, algorithms, the planned test strategy and the future scope (streaming at scale, Isolation Forest scoring, graph analytics for mule rings and production integration).

Keywords: Retrieval-Augmented Generation, Agentic AI, Fraud Investigation, Hybrid Search, Prompt-Injection Defence, Audit Trail.

Acknowledgments

It gives us great pleasure in presenting the project report on ***'Secure Hybrid Agentic RAG for Context-Aware Banking Fraud Investigation & Decision Support'****.*

We would like to take this opportunity to thank our internal guide ***Dr. Jyoti Shinde*** *for giving us all the help and guidance we needed. We are really grateful to her for her kind support. Her valuable suggestions were very helpful.*

We are also grateful to ***Dr. Omkaresh Kulkarni****, Head of the Computer Engineering Department, Dr. D. Y. Patil Institute of Technology, for his indispensable support and suggestions. We also thank our project co-ordinators* ***Dr. Swapnil Deshmukh*** *and* ***Swapnali Gaikwad*** *for their continuous direction during the project reviews.*

In the end our special thanks to the department for providing the laboratory with the needed software platforms and continuous Internet connection for our project.

Parth Pakhare
Soham Sadakal
Kaushal Pokarne
Omkar Shelar
(B.E. Computer Engg.)

Contents

1 Synopsis **1**

1.1 Project Title . . . 2
1.2 Project Option . . . 2
1.3 Internal Guide . . . 2
1.4 Sponsorship and External Guide . . . 2
1.5 Technical Keywords (As per ACM Keywords) . . . 2
1.6 Problem Statement . . . 2
1.7 Abstract . . . 3
1.8 Goals and Objectives . . . 3
1.9 Relevant Mathematics Associated with the Project . . . 3
1.10 Names of Conferences / Journals where papers can be published . . . 4
1.11 Review of Conference/Journal Papers supporting Project idea . . . 4
1.12 Plan of Project Execution . . . 5

2 Technical Keywords **6**

2.1 Area of Project . . . 7
2.2 Technical Keywords . . . 7

3 Introduction **8**

3.1 Project Idea . . . 9
3.2 Motivation of the Project . . . 9
3.3 Literature Survey . . . 9

4 Problem Definition and Scope **12**

4.1 Problem Statement . . . 13
4.1.1 Goals and objectives . . . 13
4.1.2 Statement of scope . . . 13
4.2 Major Constraints . . . 14
4.3 Methodologies of Problem Solving and Efficiency Issues . . . 14
4.4 Outcome . . . 14

4.5 Applications . . . 15
4.6 Hardware Resources Required . . . 15
4.7 Software Resources Required . . . 15

5 Project Plan **16**
5.1 Project Estimates . . . 17
5.1.1 Reconciled Estimates . . . 17
5.1.1.1 Cost Estimate . . . 17
5.1.1.2 Time Estimates . . . 17
5.1.2 Project Resources . . . 18
5.2 Risk Management with respect to NP-Hard Analysis . . . 18
5.2.1 Risk Identification . . . 18
5.2.2 Risk Analysis . . . 19
5.2.3 Overview of Risk Mitigation, Monitoring, Management . . . 20
5.3 Project Schedule . . . 21
5.3.1 Project task set . . . 21
5.3.2 Task network . . . 21
5.3.3 Timeline chart . . . 22
5.4 Team Organization . . . 25
5.4.1 Team structure . . . 25
5.4.2 Management reporting and communication . . . 25

6 Software Requirement Specification **26**
6.1 Introduction . . . 27
6.1.1 Purpose and Scope of Document . . . 27
6.1.2 Overview of Responsibilities of Developer . . . 27
6.2 Usage Scenario . . . 27
6.2.1 User profiles . . . 27
6.2.2 Use-cases . . . 28
6.2.3 Use Case View . . . 29
6.3 Data Model and Description . . . 29
6.3.1 Data Description . . . 29
6.3.2 Data objects and Relationships . . . 30
6.4 Functional Model and Description . . . 31
6.4.1 Data Flow Diagram . . . 31
6.4.1.1 Level 0 Data Flow Diagram . . . 31
6.4.1.2 Level 1 Data Flow Diagram . . . 31
6.4.2 Activity Diagram . . . 32

6.4.3 Non Functional Requirements 32
6.4.4 State Diagram 33
6.4.5 Design Constraints 33
6.4.6 Software Interface Description 33

7 Detailed Design Document 34
7.1 Introduction 35
7.2 Architectural Design 35
7.3 Data Design 35
7.3.1 Internal software data structure 35
7.3.2 Global data structure 36
7.3.3 Temporary data structure 36
7.3.4 Database description 36
7.4 Component Design 36
7.4.1 Class Diagram 36
7.4.2 Deterministic decision function 36
7.4.3 Hash-chained audit log 37
7.4.4 PII masking 37

8 Project Implementation 38
8.1 Introduction 39
8.2 Tools and Technologies Used 41
8.3 Methodologies / Algorithm Details 42
8.3.1 Algorithm 1: Fraud detection and risk aggregation 42
8.3.2 Algorithm 2: Hybrid retrieval with Reciprocal Rank Fusion . . 43
8.3.3 Algorithm 3: Bounded LangGraph investigation 44
8.4 Verification and Validation for Acceptance 45

9 Software Testing 47
9.1 Type of Testing Used 48
9.2 Test Cases and Test Results 48

10 Conclusion and Future Scope 50
10.1 Summary and Conclusion 51
10.2 Future Scope 51

11 References 52

Annexure A Laboratory Assignments on Project Analysis of Algorithmic Design 54

Annexure B Laboratory Assignments on Project Quality and Reliability Testing of Project Design **56**

Annexure C Project Planner **58**

Annexure D Reviewers' Comments of Paper Submitted **60**

Annexure E Plagiarism Report **62**

Annexure F Term-II Project Laboratory Assignments **64**

Annexure G Information of Project Group Members **66**

List of Figures

5.1 Task network 22

6.1 Use case diagram 29
6.2 Entity relationship overview 30
6.3 Level 0 data flow diagram 31
6.4 Level 1 data flow diagram 31
6.5 Activity diagram of the investigation flow 32
6.6 State transition diagram of a case 33

7.1 Architecture diagram 35
7.2 Class diagram of the main components 36

List of Tables

3.1 Literature survey 10

4.1 Methodology choices and efficiency issues 14
4.2 Hardware requirements 15

5.1 Cost estimate 17
5.2 Effort estimate for Stage 1 18
5.3 Risk table 19
5.4 Risk probability definitions [1] 19
5.5 Risk impact definitions [1] 19
5.6 Risk 2 details 20
5.7 Risk 3 details 20
5.8 Risk 4 details 21
5.9 Stage 1 weekly activity plan (W1 starts on 1 September 2026; R = review with the guide) 23
5.10 Stage 1 schedule 24
5.11 Ten-month activity plan 24
5.12 Team structure 25

6.1 User profiles 27
6.2 Use cases 28
6.3 Data objects 30

8.1 Tools and technologies 41
8.2 Rule set (initial weights; kept as data, so they can be edited) 42
8.3 Isolation Forest scores for $\psi = 256$ 42
8.4 RRF example 44

9.1 Planned test cases 49

A.1 IDEA matrix 55

CHAPTER 1

SYNOPSIS

1.1 Project Title

Secure Hybrid Agentic RAG for Context-Aware Banking Fraud Investigation & Decision Support

1.2 Project Option

Internal project.

1.3 Internal Guide

Dr. Jyoti Shinde

1.4 Sponsorship and External Guide

Not applicable. The project is not sponsored and has no external guide.

1.5 Technical Keywords (As per ACM Keywords)

1. Information systems → Information retrieval → Retrieval models and ranking
2. Computing methodologies → Artificial intelligence → Natural language processing
3. Security and privacy → Software and application security
4. Security and privacy → Human and societal aspects of security and privacy → Privacy protections
5. Applied computing → Electronic commerce → Secure online transactions

1.6 Problem Statement

To design and develop a secure, hybrid agentic RAG system that detects suspicious banking transactions, automatically investigates them by retrieving cited evidence from transaction data and internal knowledge, and supports investigators with explainable recommendations, while ensuring that the final decision is taken by a deterministic policy engine and a human investigator rather than by the language model.

1.7 Abstract

Fraud detectors flag suspicious transactions, but analysts still collect the evidence themselves. The proposed system turns a transaction into an event, scores it with deterministic rules and an Isolation Forest anomaly score, and opens a case if the risk is high. A LangGraph agent then investigates the case using read-only tools and hybrid retrieval (vector search plus BM25, merged by Reciprocal Rank Fusion) over policy and SOP files. It returns a cited summary and a typed assessment. A deterministic, versioned rule engine selects the outcome, and high-impact outcomes need human approval. PII masking, untrusted-text handling, role-based access and a hash-chained audit log secure the flow. Only synthetic data is used and all actions are simulated.

1.8 Goals and Objectives

1. To detect suspicious transactions using deterministic rules combined with an Isolation Forest anomaly model and a configurable risk aggregator.
2. To automate case investigation using a LangGraph agent restricted to predefined, read-only tools.
3. To retrieve evidence through hybrid RAG (BM25 keyword search and vector search) with source citations.
4. To obtain a bounded, typed assessment (fraud pattern, priority, evidence sufficiency) separately from the free-text summary.
5. To enforce outcomes through a deterministic policy engine with human-in-the-loop approval for high-impact cases.
6. To secure the system through PII masking, prompt-injection defences, role-based access and an append-only audit trail.

1.9 Relevant Mathematics Associated with the Project

System description. Let T be the set of transactions, C the set of cases, K the set of knowledge chunks and A the set of assessments.

- **Input:** a transaction event $t \in T$ (amount, customer, beneficiary, device, time, note) and a knowledge base K of policy chunks.
- **Output:** a case c, an assessment $a \in A$ with citations, a decision $d \in \{\text{hold}, \text{escalate}, \text{close}\}$ and a chain of audit entries.
- **Data structures:** documents in MongoDB (events, cases, rules, audit), a vector index (Chroma), an in-memory inverted index (minisearch), a small Isolation Forest model file and an in-process event queue.

- **Functions (composition of mappings):**

$$R = \text{score}(t), \qquad a = \text{investigate}(c, K), \qquad d = \text{decide}(c, a) \tag{1.1}$$

Here score : $T \to [0, 1]$, investigate : $C \times K \to A$ and decide : $C \times A \to \{\text{hold}, \text{escalate}, \text{close}\}$. Only decide produces the outcome and it contains no language model.

Risk aggregation. The rule score R_{rule}, the Isolation Forest score s and (later) a network score are combined with weights w_j, and a case is opened if the result reaches the threshold τ:

$$R = \frac{\sum_j w_j x_j}{\sum_j w_j}, \qquad \text{open a case if } R \geq \tau \tag{1.2}$$

where x_j ranges over the scores that are available.

Rank fusion. For a chunk d ranked by the vector search and the keyword search:

$$\text{RRF}(d) = \sum_{m \in \{\text{vec}, \text{bm25}\}} \frac{1}{k + \text{rank}_m(d)}, \qquad k = 60 \tag{1.3}$$

- **Success conditions:** a high-risk transaction creates a case; the agent returns a schema-valid assessment whose citations all come from retrieved chunks; the decision is reproducible for the same input; the audit chain verifies.
- **Failure conditions:** the agent exceeds its step limit (handled by a forced stop and fallback to *escalate*); an LLM output fails validation twice (fallback assessment); a citation refers to a chunk that was never retrieved (citation is removed); the audit chain does not verify.

1.10 Names of Conferences / Journals where papers can be published

- IEEE conferences on data mining, AI and applied security (for example IEEE ICDM workshops)
- ACM Workshop on Artificial Intelligence and Security (AISec)
- Conferences and workshops in IITs and central universities
- SPPU and national level student paper conferences

1.11 Review of Conference/Journal Papers supporting Project idea

Ten papers and references are reviewed in Section 3.3 of Chapter 3.

1.12 Plan of Project Execution

The ten-month plan from the approved synopsis is reproduced in Section 5.3 of Chapter 5, together with the week-by-week plan for Stage 1 (September to the start of November 2026).

CHAPTER 2

TECHNICAL KEYWORDS

2.1 Area of Project

Artificial Intelligence applied to financial security: agentic Retrieval-Augmented Generation, information retrieval and secure software design.

2.2 Technical Keywords

1. Information systems → Information retrieval
   - 1.1 Retrieval models and ranking (BM25, dense retrieval, rank fusion)
   - 1.2 Question answering and cited summarisation
2. Computing methodologies → Artificial intelligence
   - 2.1 Large language models, LLM agents, tool use
   - 2.2 Anomaly detection
3. Security and privacy
   - 3.1 Software and application security (prompt injection, retrieval poisoning)
   - 3.2 Privacy protections (PII masking), access control, audit logging

CHAPTER 3

INTRODUCTION

3.1 Project Idea

Banks flag suspicious transactions automatically, but an analyst still has to gather the evidence: transaction history, device and beneficiary history, earlier cases and the internal policies that apply. Only then can the analyst decide to *hold*, *escalate* or *close* the case. This project automates the evidence gathering while keeping people and deterministic rules in charge.

The flow is:

1. A transaction arrives as an **event**; rules and a risk score turn high-risk ones into **cases**.
2. An analyst clicks **Investigate**. An **agent** reads customer data and searches policy and SOP files using **hybrid RAG** (vector plus keyword). It returns a **cited summary** and a **typed assessment**.
3. A **deterministic decision engine** picks the outcome. The language model only recommends and never decides.
4. High-impact outcomes need **human approval**.
5. Every step is written to a **hash-chained audit log**.

All data is synthetic and all actions are simulated.

3.2 Motivation of the Project

- Digital payments through UPI, net banking and cards have grown quickly, and so have account takeover, mule accounts, beneficiary fraud and social engineering.
- Manual evidence collection is slow, differs from analyst to analyst and is difficult to audit.
- Language models can summarise heterogeneous information, but used alone they hallucinate and cannot see private bank data. Grounding them in retrieved documents with citations reduces this risk.
- Using a model inside a bank creates new risks: leakage of personal data and malicious text hidden in retrieved documents. The system should therefore separate *generation* from *decision enforcement* and keep a human in the loop.

3.3 Literature Survey

Table 3.1 summarises the ten references studied. The first five cover retrieval and agentic RAG, the sixth and tenth cover anomaly detection, and the last four cover the security of LLM applications.

Table 3.1: Literature survey

Sr.	Authors, Year	Study / Method	Key finding	Limitation
1	Lewis et al., 2020 [2]	Retrieval-augmented generation: dense retrieval with a seq2seq generator	Grounding generation in retrieved passages improves factual accuracy	Single-shot retrieval; no reasoning over structured or relational data
2	Yao et al., 2023 [3]	ReAct: interleaved reasoning traces and tool actions	Tool use during reasoning reduces hallucination	Free-running loop; no control over which actions may run
3	Asai et al., 2024 [4]	Self-RAG: adaptive retrieval with reflection tokens	Model decides when to retrieve and critiques its own output	Needs fine-tuning; not tested on private, access-controlled data
4	Edge et al., 2024 [5]	Graph RAG: knowledge graph with community summaries	Answers global, relationship-level questions better than vector RAG	High indexing cost; security of retrieved content not addressed
5	Singh et al., 2025 [6]	Survey of single- and multi-agent RAG	Classifies planning, routing and graph-based workflows	Governance, auditability and human oversight remain open
6	Hilal et al., 2022 [7]	Review of statistical, ML and deep anomaly detection for fraud	Anomaly detection works on unlabelled fraud data	Detectors flag alerts but do not explain them
7	Greshake et al., 2023 [8]	Indirect prompt injection attacks on LLM-integrated apps	Instructions hidden in retrieved content can hijack the model	Shows the need to treat retrieved text as untrusted data
8	Inan et al., 2023 [9]	Llama Guard: LLM as a bounded-label safety classifier	A model limited to fixed categories gives checkable output	Built for content safety, not domain decisions such as fraud typing
9	Zou et al., 2025 [10]	PoisonedRAG: knowledge-corruption attacks	A few poisoned texts in the knowledge base can steer the answer	Motivates controlled, versioned policy files and output checks
10	Liu et al., 2008 [11]	Isolation Forest: anomalies need fewer random splits to isolate	Fast, works without labels, suitable for transactions	Gives a score only, no explanation or context

Gaps identified

- Fraud detection systems generate alerts but offer little explanation, context or evidence to support the investigation.
- RAG and agentic systems are rarely designed for private, role-restricted financial data with PII protection.
- Most agentic frameworks let the model act directly; generative output and de-

cision enforcement are not separated.

- Defences against prompt injection and retrieval poisoning are seldom built into the investigation workflow.
- Bounded typed outputs, human review and complete audit trails are not combined in one system.

CHAPTER 4

PROBLEM DEFINITION AND SCOPE

4.1 Problem Statement

Fraud analysts spend a large share of their time collecting evidence for alerts raised by detectors. The evidence is spread over transaction records, customer and device history, and policy documents, and the resulting decisions are hard to reproduce and audit. The problem is to build a system that collects and cites this evidence automatically, recommends an action, and still guarantees that a deterministic rule engine and a human, not the language model, take the final decision.

4.1.1 Goals and objectives

- **Goal:** reduce the effort of gathering evidence for a flagged transaction while keeping every decision explainable and auditable.
- **Input:** a transaction event (JSON, validated by a schema), policy files (PDF, Markdown, TXT) uploaded by an authorised user, and an analyst's *Investigate* request.
- **Output:** a case record, a summary with citations (chunk identifier, source file, page), a typed assessment, a decision with rule version and reasons, an approval record when needed, and audit entries.

4.1.2 Statement of scope

The system will:

- ingest transaction events, score them and create cases automatically;
- provide create/read/update/delete screens for customers, transactions and cases, and an events feed;
- let authorised users upload, version and archive policy files, and search them with a hybrid retriever;
- investigate a case with a bounded agent and show the summary, assessment and citations;
- decide with a versioned rule engine, send high-impact cases to a supervisor, and run simulated actions;
- mask personal data before every model call, treat external text as untrusted, apply role-based access and keep a hash-chained audit log.

The system will not: use real customer data, connect to any real banking system, stream at production scale, fine-tune models, or let the language model decide an outcome.

Size and bounds. The prototype is designed for about 50 synthetic customers, 500 transactions and 30 to 50 policy documents. Inputs are validated with schemas (types, ranges, enumerations); invalid input is rejected with an error message.

4.2 Major Constraints

- Only synthetic data may be used; no real personal or banking data.
- The chat model is reached through the OpenRouter API; rate limits and network failures must not stop the demo (a built-in fake model is used as a fallback).
- Embeddings run locally (about 100 MB model download on first use).
- The language model must never take the final decision, and every state change must be audited.

4.3 Methodologies of Problem Solving and Efficiency Issues

Concern	**Alternative considered**	**Approach chosen and reason**
Detection	Rules only; ML only	Weighted rules plus an Isolation Forest: the rules are explainable, and the forest is unsupervised, needs no labelled data and is small enough to run in-process in Node.js.
Retrieval	Vector only; keyword only	Hybrid with Reciprocal Rank Fusion: keywords catch exact codes and terms, vectors catch meaning, and fusion needs no score calibration.
Agent	Free-running agent loop	Graph with a step cap: always terminates and is easy to trace.
Decision	Let the LLM decide	Deterministic versioned rules: reproducible and auditable.
Events	Kafka	In-process event emitter behind an interface: enough for Stage 1, replaceable later.
Efficiency	—	Local embeddings are cached; the keyword index is in memory; investigations run asynchronously so the UI never blocks.

Table 4.1: Methodology choices and efficiency issues

4.4 Outcome

- A working prototype in which a suspicious synthetic transaction becomes a case, is investigated with cited evidence, receives a deterministic decision and, if needed, a supervisor's approval, with a verifiable audit trail.

- A measured comparison of vector-only, keyword-only and hybrid retrieval on a small hand-written question set.
- Security tests showing that poisoned policy text and injected transaction notes do not change the decision and that no raw PII reaches the model.

4.5 Applications

Fraud operations teams in banks, payment service providers and NBFCs can investigate alerts faster and more consistently. Compliance and audit teams can trace each decision to its evidence, policy version and reviewer. The same design can be extended to anti-money-laundering review and insurance-claim fraud.

4.6 Hardware Resources Required

Sr. No.	Parameter	Minimum requirement	Justification
1	CPU	4 cores, 2.5 GHz	Runs the API, UI, MongoDB and the local embedding model
2	RAM	8 GB	MongoDB and Chroma containers, Node.js processes and the embedding model
3	Disk	10 GB free	Docker images, embedding model (about 100 MB), data volumes
4	Network	Stable broadband	OpenRouter API calls and first-time downloads

Table 4.2: Hardware requirements

4.7 Software Resources Required

Platform:

1. Operating system: Windows 10/11, Linux or macOS with Docker Desktop
2. IDE: Visual Studio Code
3. Programming language: JavaScript (Node.js 20+, ES modules)
4. Database and search: MongoDB, Chroma, minisearch
5. Frameworks: Express, React with Vite, TanStack Query, Tailwind CSS, LangChain.js, LangGraph.js
6. Other: Git and GitHub, GitHub Actions, OpenRouter account

CHAPTER 5

PROJECT PLAN

5.1 Project Estimates

The waterfall model is followed: requirements, design, implementation, testing, documentation. Stage 1 runs for ten weeks, from September to the start of November 2026, with four developers working part-time alongside their academic load (about three productive days per week each); the complete project runs over ten months. Reviews with the internal guide are scheduled at the end of Weeks 4, 7 and 10.

5.1.1 Reconciled Estimates

5.1.1.1 Cost Estimate

All software used is free or open source. The only recurring cost is the OpenRouter API usage for the chat model, which is kept small by using a low-cost model and a built-in fake model for tests and demos.

Item	**Estimated cost**
Software licences (Node.js, MongoDB, Chroma, React, VS Code)	Rs. 0 (open source)
OpenRouter API credits (development and demo)	**[TODO: team to fix a budget]**
Hardware	Existing student laptops

Table 5.1: Cost estimate

5.1.1.2 Time Estimates

Stage 1 spans 10 weeks (1 September – 7 November 2026). With four developers at about three productive days per week, the available effort is $4 \times 10 \times 3 = 120$ person-days.

Activity	Weeks	Persons	Effort (person-days)
Planning: problem definition, literature survey, requirement gathering, learning the stack	1–2	4	24
Design: architecture, data model, threat modelling, API contracts	3–4	4	24
Setup: repository, Docker, CI, frozen contracts, stubs	5	4	4
Implementation of own module: core backend (P1), hybrid RAG (P2), agent and UI (P3), decision and security (P4)	5–8	1 each	12 each (48)
Integration, testing, demo, reviews and report	8–10	4	20
Total for Stage 1	**10**		**120**

Table 5.2: Effort estimate for Stage 1

5.1.2 Project Resources

People: four student developers and one internal guide. **Hardware:** laptops as in Table 4.2. **Software:** the tools listed in Chapter 4 and Chapter 8. **Other:** GitHub repository with protected `main` branch, GitHub Actions for continuous integration, and an OpenRouter account. Work is divided so that each developer owns separate folders, which avoids edit conflicts; components communicate through frozen data contracts and stub functions.

5.2 Risk Management with respect to NP-Hard Analysis

The core computations of the system are polynomial: scoring a transaction is $O(n)$ in the number of rules, fusing two ranked lists of length N is $O(N \log N)$, and the agent is bounded by a step limit, so every investigation finishes in a bounded number of steps. Choosing an *optimal* set of rules and weights for a given data set can be hard in general (it is a combinatorial selection problem), so the project uses fixed, expert-written rules, a simple weighted sum and a threshold that is tuned by a sweep, i.e. a heuristic rather than an exact optimum. This section lists the project risks and how they are managed.

5.2.1 Risk Identification

Risks were identified by reviewing the scope document, requirements and schedule, and by answering the standard risk questionnaire [1] (commitment of the guide, stability of requirements, skills of the team, team size and so on). Table 5.3 lists them.

5.2.2 Risk Analysis

ID	Risk description	Probability	Impact		
			Schedule	Quality	Overall
1	Chroma or Docker does not run on a team member's laptop	Medium	Medium	Low	Medium
2	LLM provider rate limit or network failure during a demo	Medium	Low	Medium	Medium
3	Prompt injection or a poisoned policy file changes behaviour	Low	Low	High	High
4	Late integration or data-contract drift between owners	Medium	High	Medium	High
5	Model output is invalid or not supported by evidence (hallucination)	Medium	Low	High	High

Table 5.3: Risk table

Probability	Value	Description
High	Probability of occurrence is	$> 75\%$
Medium	Probability of occurrence is	26–75%
Low	Probability of occurrence is	$< 25\%$

Table 5.4: Risk probability definitions [1]

Impact	Value	Description
Very high	$> 10\%$	Schedule impact or unacceptable quality
High	5–10%	Schedule impact or some parts of the project have low quality
Medium	$< 5\%$	Schedule impact or barely noticeable degradation in quality
Low	—	Impact on schedule or quality can be absorbed

Table 5.5: Risk impact definitions [1]

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Risk ID	2
Risk description	The LLM provider (OpenRouter) is rate limited or unreachable during development or demonstration
Category	Technology
Source	Early development and testing
Probability / Impact	Medium / Medium
Response	Mitigate
Strategy	A deterministic fake LLM (`USE_FAKE_LLM=true`) returns schema-valid output so tests and demos run offline; keep cached sample investigations
Risk status	Identified

Table 5.6: Risk 2 details

Risk ID	3
Risk description	Prompt injection or a poisoned policy file tries to change the outcome
Category	Security
Source	Literature survey [8, 10]
Probability / Impact	Low / High
Response	Mitigate
Strategy	Untrusted text is delimited and filtered; the LLM never decides; the decision engine is plain code; security tests with a poisoned document and an injected note must pass
Risk status	Identified

Table 5.7: Risk 3 details

Risk ID	4
Risk description	Late integration or schema drift between the four owners
Category	Requirements / Process
Source	Project schedule review
Probability / Impact	Medium / High
Response	Mitigate
Strategy	Data contracts drafted during design and frozen in Week 5; stubs for every public function; mini end-to-end check with stubs at the end of Week 5; small pull requests with cross-area review
Risk status	Identified

Table 5.8: Risk 4 details

5.3 Project Schedule

5.3.1 Project task set

Major tasks in Stage 1 are:

- Task 1: Freeze data contracts, set up repository, Docker services and CI.
- Task 2: Build the event system, risk scoring, case creation and the CRUD screens.
- Task 3: Build the policy ingestion pipeline and the hybrid search.
- Task 4: Build the agent, the fake and real LLM clients and the Investigate screen.
- Task 5: Build the decision engine, approvals, security layer and audit log.
- Task 6: Integrate, test (including security tests), prepare demo scenarios.

5.3.2 Task network

Figure 5.1 shows the dependencies between the tasks.

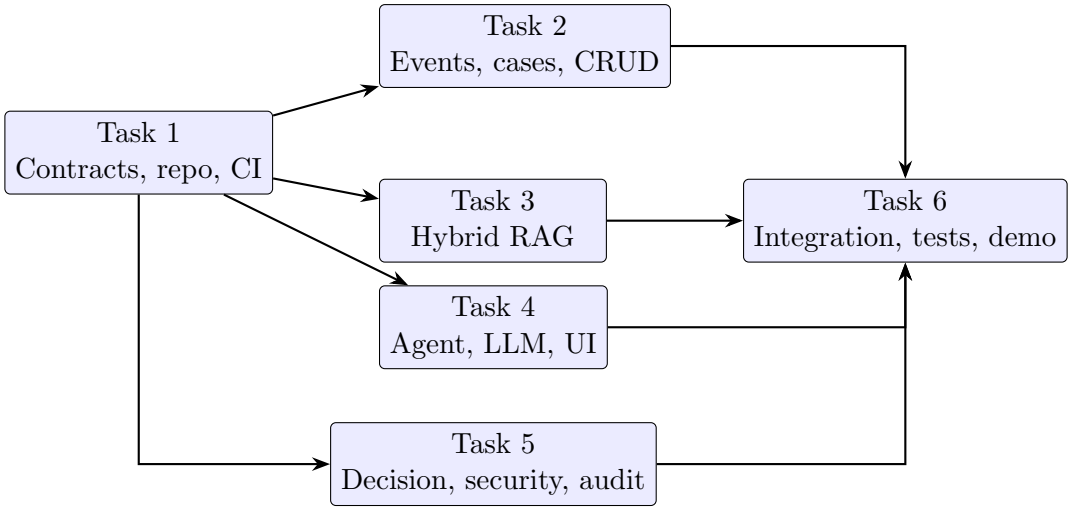


Figure 5.1: Task network

5.3.3 Timeline chart

Stage 1 Gantt chart (weekly, 1 September – 7 November 2026).

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Problem definition and literature survey										
Requirement gathering and learning the stack										
Architecture, data model, threat modelling										
API and data contracts (draft, then frozen)										
Repository, Docker, CI and stubs										
Synthetic data, events, rules, Isolation Forest										
Case, transaction and audit services										
Policy ingestion and hybrid retrieval										
LangGraph agent and LLM client										
Decision engine, approvals, security										
React console and Investigate screen										
Integration and simulator scenarios										
Unit, integration and end-to-end testing										
Security tests and retrieval evaluation										
Demo, README and freeze										
Report writing and finalisation										
Guide review				R1			R2			R3

Table 5.9: Stage 1 weekly activity plan (W1 starts on 1 September 2026; R = review with the guide)

Weeks	Dates (2026)	Work and milestone
1–2	1–12 Sep	**Planning only.** Problem definition, literature survey, requirement gathering; learning the stack (RAG, LangGraph, MongoDB, Isolation Forest)
3–4	14–26 Sep	**Design only.** Architecture, data model, threat modelling, draft of API and data contracts; no production code. **Review R1** (plan and design)
5	28 Sep – 3 Oct	**Setup.** Repository, Docker services, CI; contracts frozen; stubs for every public function; mini end-to-end check with stubs; first spikes of each module
6–7	5–17 Oct	**Implementation.** Each owner builds their module against stubs; synthetic data; Isolation Forest trained; first screens of the console. **Review R2** (working modules)
8	19–24 Oct	**Integration.** Stubs replaced by real modules; transaction → case → investigate → decide → approve → audit; tag `v0.1`
9	26–31 Oct	**Testing.** Security tests, retrieval and detection evaluation, bug fixing
10	2–7 Nov	**Demo.** Demo scenarios, README, report finalisation, freeze and tag `v1.0-phase1`. **Review R3** (demonstration)

Table 5.10: Stage 1 schedule

Overall plan (ten months, from September 2026). Stage 1 ends at the start of November; the later stages extend the same code base as outlined in the future scope (Chapter 10).

Activity	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Stage 1: planning, requirements, architecture and design										
Stage 1: setup, core services, hybrid retrieval, agent, decision engine, UI										
Stage 1: integration, testing, demo (tag `v1.0-phase1`)										
Streaming and workers, richer detection										
Graph analytics and RAG improvements										
Agent improvements and assessment providers										
Security hardening and red-team tests										
Evaluation and ablation experiments										
Documentation and research paper										
Report writing and finalisation										

Table 5.11: Ten-month activity plan

5.4 Team Organization

5.4.1 Team structure

Role	Area	Main responsibility
P1	Core backend, events and CRUD	Express base app, data models, event system, risk scoring, seed and simulator, case and transaction screens
P2	Hybrid RAG and policy files	Ingestion, chunking, local embeddings, vector and keyword indexes, Reciprocal Rank Fusion, policy versioning, retrieval evaluation
P3	Agent, OpenRouter and Investigate UI	LLM client with fake mode, read-only tools, LangGraph flow, typed assessment, citation checks, Investigate screen
P4	Decision engine, security and infra	Deterministic rules and `decide()`, approvals, PII masking, untrusted-text handling, RBAC, audit chain, Docker and CI

Table 5.12: Team structure

5.4.2 Management reporting and communication

A 10-minute daily sync (what I finished, what I will do, what blocks me); work on short-lived `feat/` branches merged to `develop` by small pull requests with one reviewer from another area; weekly progress review with the internal guide as per the project assessment schedule.

CHAPTER 6

SOFTWARE REQUIREMENT SPECIFICATION

6.1 Introduction

6.1.1 Purpose and Scope of Document

This chapter states what the system must do (functional requirements) and the qualities it must have (non-functional requirements), in terms of users, use cases, data and processes. It does not describe implementation detail; that is in Chapters 7 and 8.

6.1.2 Overview of Responsibilities of Developer

The developers analyse requirements, design the architecture and data contracts, implement the modules, write the tests, prepare synthetic data and demo scenarios, and document the work.

6.2 Usage Scenario

6.2.1 User profiles

Actor	**Description**
Fraud analyst	Reviews cases, clicks Investigate, reads the evidence and the recommendation
Supervisor	Approves or rejects high-impact decisions, reviews rules and the audit log
Administrator	Manages policy files, rules and user roles
Transaction source (simulator)	Sends synthetic transaction events to the system

Table 6.1: User profiles

6.2.2 Use-cases

Sr.	Use case	Description	Actors	Assumption
1	Submit transaction	A transaction event is received, scored and may create a case	Source	Event matches the schema
2	Manage cases	Create, view, update and delete cases and transactions	Analyst	User is logged in
3	Investigate case	Agent gathers evidence and returns cited summary and assessment	Analyst	Policy files are indexed
4	Review evidence	Open citations to see the source passage and page	Analyst	Investigation completed
5	Approve or reject	Decide on a pending high-impact outcome	Supervisor	Case is in pending approval
6	Manage policy files	Upload, version, archive and reindex files	Admin	File is PDF, MD or TXT
7	Manage rules	View rules and create a new version of a rule	Supervisor, Admin	Old versions are kept
8	View audit log	Browse entries and verify the hash chain	Supervisor, Admin	Audit is append-only

Table 6.2: Use cases

6.2.3 Use Case View

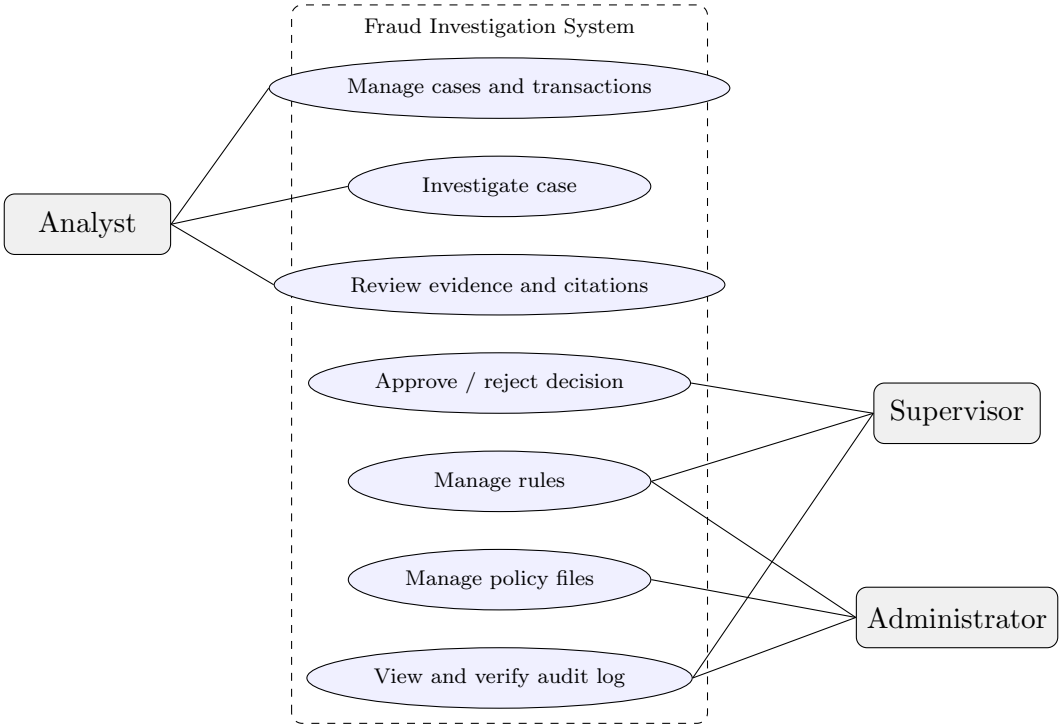


Figure 6.1: Use case diagram

6.3 Data Model and Description

6.3.1 Data Description

Data objects are stored as MongoDB documents and validated by shared Zod schemas.

Object	Main attributes
Customer	id, name, segment, accountAgeDays, riskTier, devices[], knownBeneficiaries[]
Transaction	id, customerId, amount, currency, channel (UPI / NETBANKING / CARD), beneficiaryId, deviceId, ip, note, timestamp, riskScore, flags[]
Event	id, type, payload, source, createdAt, processedAt
Case	id, transactionIds[], customerId, status, priority, riskScore, assessment, decision, summary
PolicyFile	id, name, version, mimeType, status (active / archived), chunkCount
Chunk	id (citation id), policyFileId, version, page, text, embeddingModel
Assessment	fraudPattern, priority, evidenceSufficiency, recommendedAction, summary, citations[], confidence
Rule	id, version, name, condition, outcome, requiresApproval, active
Decision	caseId, outcome, requiresApproval, ruleId, ruleVersion, reasons[], approvedBy
AuditEntry	id, ts, actor, action, entityType, entityId, details, prevHash, hash

Table 6.3: Data objects

6.3.2 Data objects and Relationships

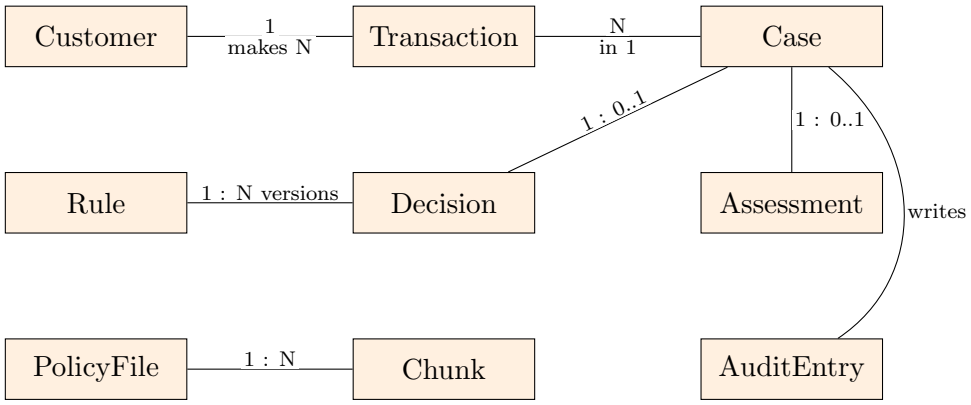


Figure 6.2: Entity relationship overview

6.4 Functional Model and Description

6.4.1 Data Flow Diagram

6.4.1.1 Level 0 Data Flow Diagram

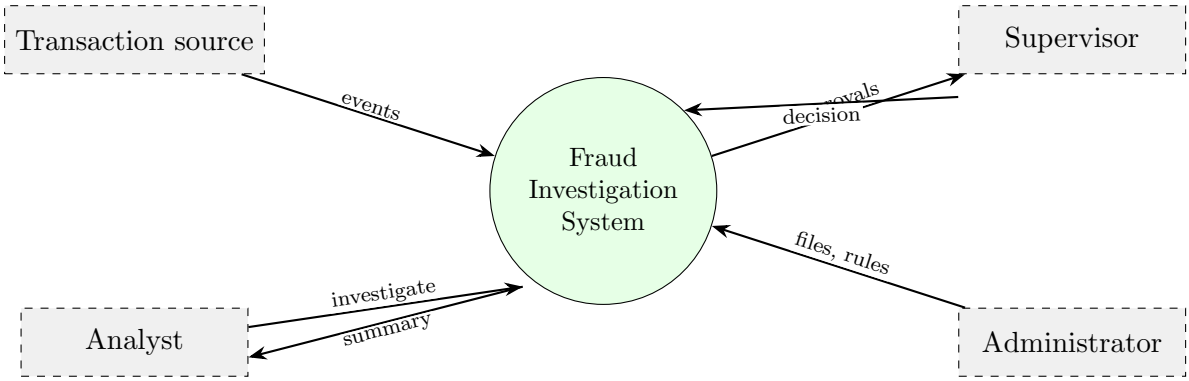


Figure 6.3: Level 0 data flow diagram

6.4.1.2 Level 1 Data Flow Diagram

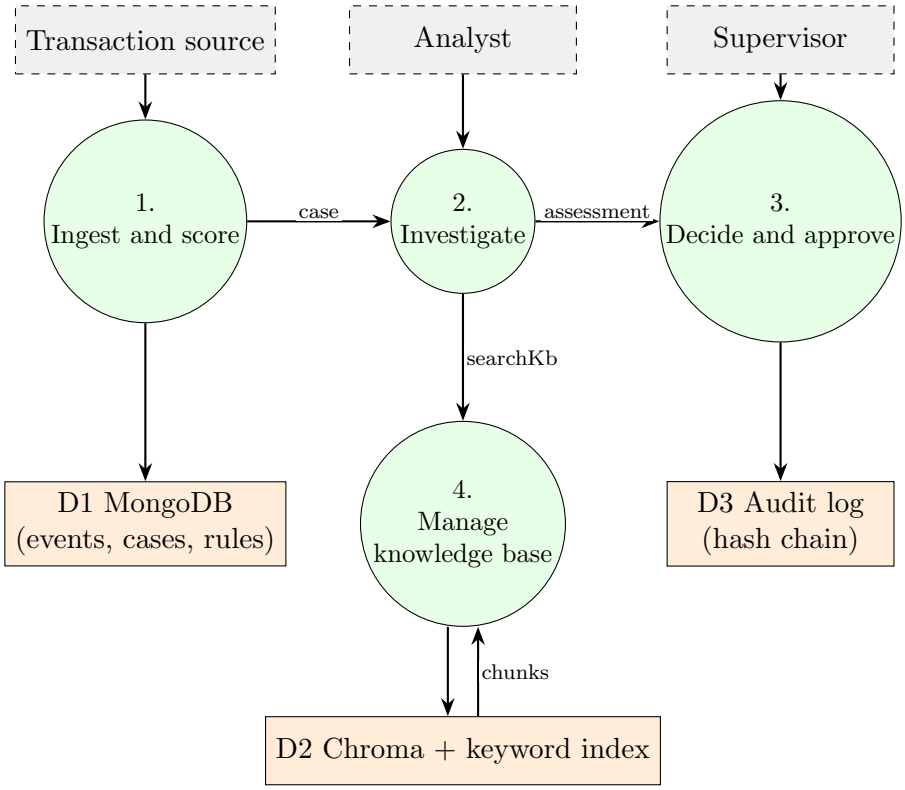


Figure 6.4: Level 1 data flow diagram

6.4.2 Activity Diagram

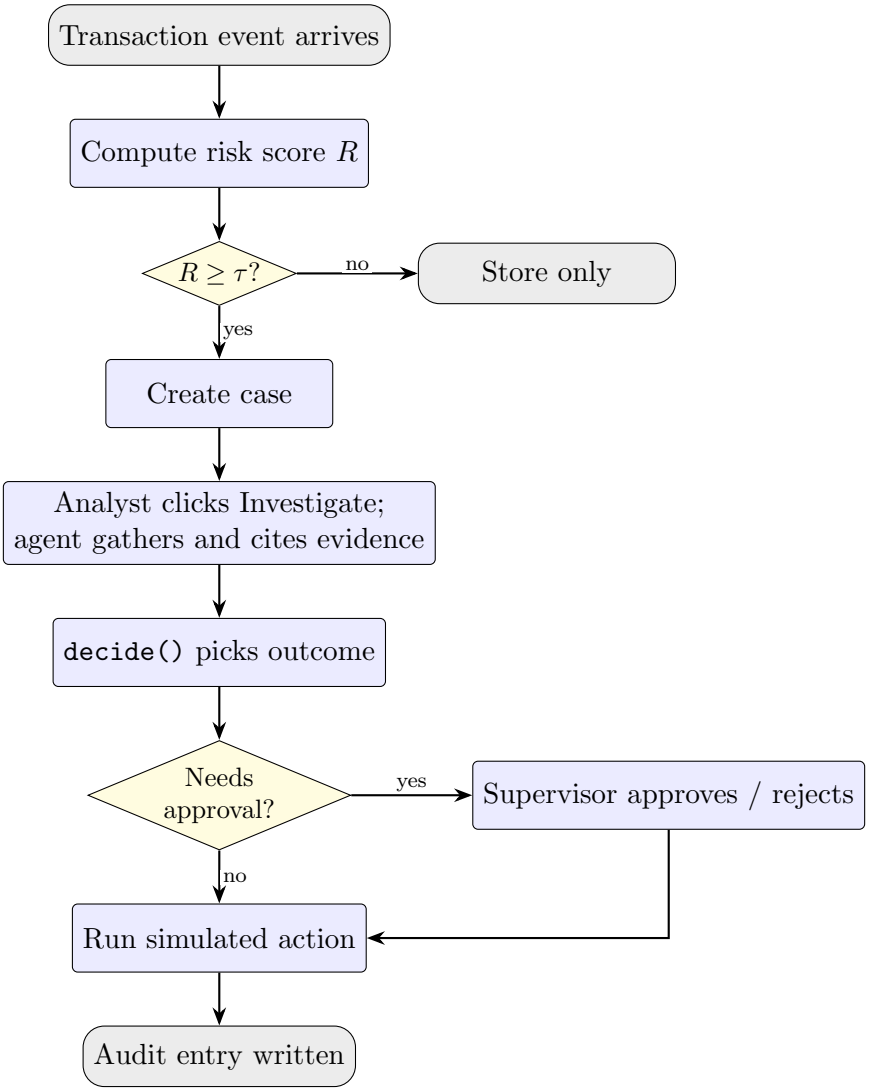


Figure 6.5: Activity diagram of the investigation flow

6.4.3 Non Functional Requirements

- **Interface requirements:** web interface in a modern browser; JSON REST API; every request body validated by a schema; errors returned in a common format.
- **Performance requirements:** list screens respond in under 2 seconds on the prototype data set; an investigation runs asynchronously and finishes within the step cap (the time depends on the LLM, so the UI shows live status).
- **Security:** PII masked before any model call; retrieved text and notes treated as untrusted; role-based access (analyst, supervisor, admin); append-only hash-chained audit log; no secrets in the repository.
- **Reliability and availability:** the built-in fake model keeps the system usable when the LLM provider is down; the investigation always terminates.

- **Modifiability and portability:** the event system and vector store sit behind interfaces so a message broker or another vector database can replace them; Docker Compose starts the dependencies on any machine.
- **Testability:** every module has unit tests; the fake model makes agent tests deterministic.
- **Usability:** each result shows the recommendation, the rule that fired and the cited evidence on one screen.

6.4.4 State Diagram

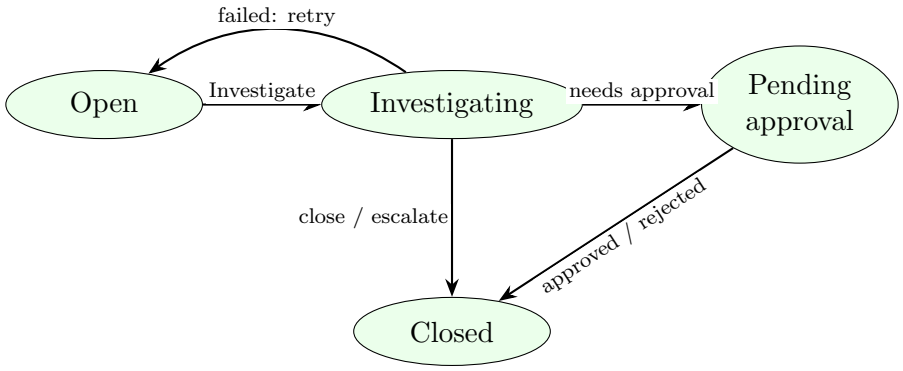


Figure 6.6: State transition diagram of a case

6.4.5 Design Constraints

Synthetic data only; the language model cannot decide; all actions are simulated; each folder has a single owner and the data contracts are frozen after Week 5.

6.4.6 Software Interface Description

- **User interface:** React single-page application calling the REST API under `/api/v1`.
- **LLM interface:** OpenAI-compatible chat API of OpenRouter, reached through LangChain.js.
- **Database interface:** Mongoose for MongoDB; the Chroma client for vectors.
- **Files:** policy documents (PDF, Markdown, text) uploaded through a multipart request.

CHAPTER 7

DETAILED DESIGN DOCUMENT

7.1 Introduction

This chapter describes the structure of the system and the data it keeps. It follows the architecture in which the user interface talks to one API, the API is split into modules owned by different developers, and the only components allowed to produce an outcome are the deterministic ones.

7.2 Architectural Design

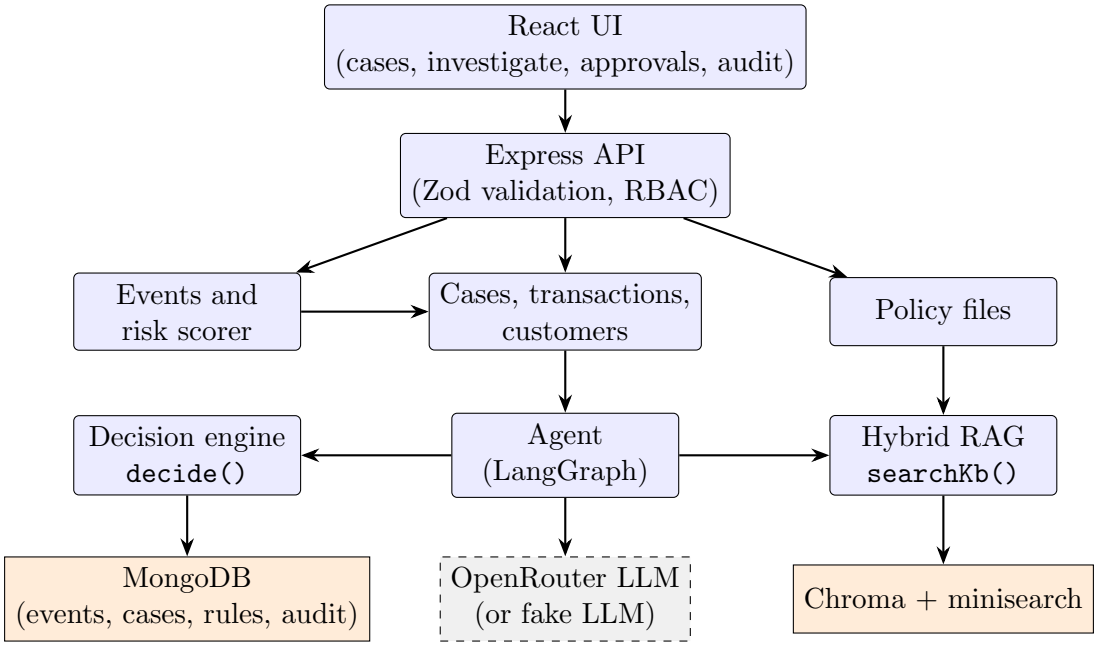


Figure 7.1: Architecture diagram

Package view. The repository is a single monorepo with npm workspaces: `apps/api` (modules `events`, `cases`, `transactions`, `customers`, `policies`, `rag`, `agent`, `decision`, `security`), `apps/web` (pages, components, API client) and `packages/shared` (Zod schemas and event types). MongoDB and Chroma run in Docker; the API and UI run locally during development.

7.3 Data Design

7.3.1 Internal software data structure

Objects passed between components are validated by the shared schemas: the `Event` (producer to consumers), the `Chunk` list returned by `searchKb` (retrieval to agent), the `Assessment` (agent to decision engine) and the `Decision` (decision engine to approval queue).

7.3.2 Global data structure

The Zod-validated configuration (risk threshold, step cap, model names, URLs) and the shared constants (event types, status values) are available to all modules.

7.3.3 Temporary data structure

A per-investigation record of retrieved chunk identifiers (used to check citations), a reversible PII token map kept in memory only for one run, and the in-memory keyword index rebuilt from MongoDB at start-up.

7.3.4 Database description

MongoDB collections: `events`, `customers`, `transactions`, `cases`, `policyfiles`, `chunks`, `rules`, `audit`. Chroma collection: policy chunk vectors with metadata (file, version, page, status).

7.4 Component Design

7.4.1 Class Diagram

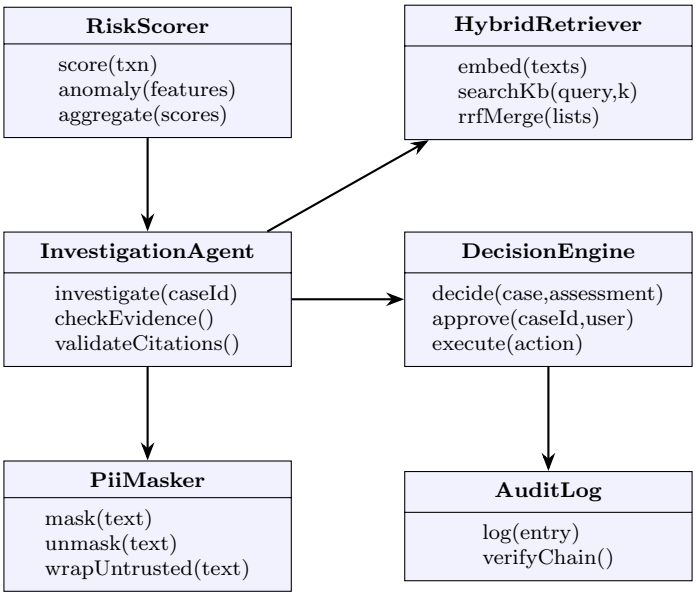


Figure 7.2: Class diagram of the main components

7.4.2 Deterministic decision function

The decision engine holds a list of versioned rules $r_1, \ldots, r_m$ in priority order. Each rule has a condition over the risk score, the fraud pattern, the priority, the evidence

sufficiency and the amount. The decision is

$$\text{decide}(c, a) = \begin{cases} \text{outcome}(r_j) & j = \min\{\, i : \text{cond}_i(c, a) \text{ is true} \,\} \\ \text{escalate} & \text{if no rule matches} \end{cases} \tag{7.1}$$

The same input always gives the same output, the rule identifier and version are stored with the decision, and no language model is called. If the matching rule has the flag *requiresApproval*, the case moves to `pending_approval` until a supervisor or administrator approves or rejects it.

7.4.3 Hash-chained audit log

Each audit entry e_n stores the hash of the previous entry:

$$h_n = \text{SHA256}\big(h_{n-1} \,\|\, \text{canon}(e_n)\big) \tag{7.2}$$

where canon writes the entry as JSON with a fixed key order. To verify the log, the hashes are recomputed from the first entry; if one entry was edited or removed, its hash and all later hashes no longer match, so the change is detected.

7.4.4 PII masking

Before a model call, names, account numbers, phone numbers, emails and IP addresses are replaced by stable tokens such as `[ACCT_1]`. The token map exists only in memory for that run, and original values are restored only for display in the user interface.

CHAPTER 8

PROJECT IMPLEMENTATION

8.1 Introduction

Stage 1 builds the complete investigation flow as a working prototype. This chapter lists the tools and then explains the three central algorithms in simple terms: risk scoring, hybrid retrieval and the bounded agent. The rule engine and the audit chain are described with their equations in Chapter 7.

8.2 Tools and Technologies Used

Layer	Tool	Use in the project
Runtime	Node.js 20+, JavaScript (ES modules)	One language for API, workers, scripts and UI
Backend	Express, Zod	REST API; schema validation of requests, events and LLM output
Database	MongoDB with Mongoose	Events, cases, rules, audit log, file metadata
Vector store	Chroma	Stores chunk embeddings and answers similarity queries
Keyword search	minisearch	In-memory BM25-style

8.3 Methodologies / Algorithm Details

8.3.1 Algorithm 1: Fraud detection and risk aggregation

Idea. Every transaction gets a risk number between 0 and 1. It combines two views: (i) *rules* written by experts, which are easy to explain, and (ii) an *Isolation Forest*, which finds transactions that look unusual compared with normal ones without needing fraud labels.

Rule score. Six rules each give a flag $f_i \in [0, 1]$ and have a weight w_i with $\sum w_i = 1$:

$$R_{\text{rule}} = \sum_{i=1}^{6} w_i f_i \tag{8.1}$$

i	Rule	w_i	Flag f_i
1	New beneficiary and high amount	0.25	1 if both are true, else 0
2	New device	0.20	1 or 0
3	Velocity (many transactions in 10 minutes)	0.15	1 or 0
4	Amount compared with customer average μ_c	0.25	$\min\big(1,\ x/(5\mu_c)\big)$
5	Odd hour of the day	0.10	1 or 0
6	IP location mismatch	0.05	1 or 0

Table 8.2: Rule set (initial weights; kept as data, so they can be edited)

Isolation Forest. The key idea: a strange point is easy to separate from the others, so it needs only a few random splits. The model is a set of $T = 100$ small random trees. Each tree is built from a random sample of $\psi = 256$ points: pick a random feature, pick a random split value between its minimum and maximum, and repeat until the point is alone or the depth reaches $\lceil \log_2 \psi \rceil$. For a new point x the *path length* $h(x)$ is the number of splits needed to isolate it, averaged over all trees as $E[h(x)]$. The anomaly score is [11]

$$s(x) = 2^{-E[h(x)]/c(\psi)}, \qquad c(\psi) = 2\big(\ln(\psi - 1) + 0.5772\big) - \frac{2(\psi - 1)}{\psi} \tag{8.2}$$

where $c(\psi)$ is the average path length for ψ points, used to normalise. A short path gives a score close to 1 (anomalous); an average path gives about 0.5. For $\psi = 256$ we get $c \approx 10.2$:

$E[h(x)]$	$s(x)$	Meaning
3	0.82	Isolated very quickly: very unusual
5	0.71	Isolated quickly: unusual
10	0.51	Average path: normal

Table 8.3: Isolation Forest scores for $\psi = 256$

The features come from the transaction and the customer history (amount ratio, velocity in time windows, new-device and new-beneficiary flags, hour of day). The model is trained offline on synthetic data with a fixed random seed, saved as a small versioned JSON file and loaded once. If the file is missing or its feature list does not match, the system falls back to rules only.

Aggregation. The available scores are combined with configurable weights (for example rule 0.4, anomaly 0.4, network 0.2 in the future, where the network score is absent in Stage 1):

$$R = \frac{\sum_{j \in \text{available}} w_j x_j}{\sum_{j \in \text{available}} w_j} \tag{8.3}$$

Worked example. A new device (1, weight 0.20), a new beneficiary with a high amount (1, 0.25), no velocity alert (0), an amount of $4\mu_c$ ($f_4 = 0.8$, 0.25), an odd hour (1, 0.10) and no IP mismatch (0) give $R_{\text{rule}} = 0.20{+}0.25{+}0{+}0.25{\times}0.8{+}0.10{+}0 = 0.75$. If the forest gives $s = 0.70$, then $R = (0.4 \times 0.75 + 0.4 \times 0.70)/(0.4 + 0.4) = 0.725 \geq \tau = 0.7$, so a case is created.

Algorithm 1: Risk scoring and case creation

Input: transaction t, customer history, rule set, Isolation Forest model M (optional), weights, threshold τ
Output: risk breakdown R; a case if $R \geq \tau$
for *each rule* i **do**
  compute flag f_i from t and the history
$R_{\text{rule}} \leftarrow \sum_i w_i f_i$
if *M is available and its feature list matches* **then**
  $s \leftarrow$ Isolation Forest score of the feature vector of t
else
  $s \leftarrow$ **absent** (rules-only mode, flagged as degraded)
$R \leftarrow$ weighted average of the available scores
store R_{rule}, s and R with the transaction; publish `txn.scored`
if $R \geq \tau$ **then**
  create a case; publish `case.created`; write an audit entry

8.3.2 Algorithm 2: Hybrid retrieval with Reciprocal Rank Fusion

Idea. Two searches complement each other. *Keyword search* (BM25) finds exact words such as a policy code or a country name. *Vector search* finds passages with similar meaning even with different words. Their score scales differ, so instead of mixing scores we mix *ranks*.

Vector similarity. The query and each chunk are converted to vectors q and d by the same embedding model, and compared with the cosine:

$$\cos(q, d) = \frac{q \cdot d}{\|q\| \, \|d\|} \tag{8.4}$$

Keyword score (BM25). For each query term t in chunk d [12]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \cdot \frac{\mathrm{tf}(t,d)\,(k_1+1)}{\mathrm{tf}(t,d) + k_1\big(1 - b + b\,|d|/\mathrm{avgdl}\big)} \tag{8.5}$$

Here tf is how often the term occurs in the chunk, IDF gives rare terms more weight, and $|d|/\mathrm{avgdl}$ is the chunk length compared with the average. We use the usual defaults of the library.

Reciprocal Rank Fusion. With the two ranked lists, each chunk receives

$$\mathrm{RRF}(d) = \sum_{m \in \{\mathrm{vec},\,\mathrm{bm25}\}} \frac{1}{k + \mathrm{rank}_m(d)}, \qquad k = 60 \tag{8.6}$$

A chunk missing from a list adds 0 for that list. The constant k keeps the first ranks from dominating [13].

Worked example. Three chunks A, B, C are ranked as follows.

Chunk	Vector rank	BM25 rank	RRF
A	1	3	$1/61 + 1/63 = 0.03227$
B	2	1	$1/62 + 1/61 = 0.03252$
C	3	2	$1/63 + 1/62 = 0.03200$

Table 8.4: RRF example

Chunk B is returned first, although it was not first in the vector list, because both searches rank it high.

Algorithm 2: `searchKb(query, filters, k)`

Input: query text, filters (active versions only), number of results k, list size $N = 20$
Output: top-k chunks with source file and page
$q \leftarrow$ embed(query)
$V \leftarrow$ top-N chunks by cosine similarity (Chroma)
$B \leftarrow$ top-N chunks by BM25 (minisearch)
remove chunks that are archived or do not match the filters
for *each chunk* d *in* $V \cup B$ **do**
  $\mathrm{RRF}(d) \leftarrow \sum_m 1/(60 + \mathrm{rank}_m(d))$
sort by RRF in descending order
return the first k chunks with their `chunkId`, source and page

8.3.3 Algorithm 3: Bounded LangGraph investigation

Idea. The agent is a small graph, not a free loop: *gather context* → *retrieve* → *check evidence* → (retrieve again or *assess*) → *summarise*. A counter s is increased at every retrieval and the loop stops when $s = T_{\max}$ (`AGENT_MAX_STEPS`, default 6), so the agent *always terminates*. It can only call read-only tools (customer, transaction history, device, `searchKb`).

Safe use of text. Before any model call, personal data is masked and all retrieved text and free-text fields are wrapped in delimiters and presented to the model as data, not instructions.

Typed output with a fallback. The assessment must pass the Zod schema. If it fails, the agent retries once; if it fails again it returns the safe fallback *(unknown, insufficient, escalate)*.

Citation check. Let C be the chunk identifiers cited by the model and R the identifiers actually returned by `searchKb` in this run. Only valid citations are kept:

$$C_{\text{valid}} = C \cap R \tag{8.7}$$

Citations outside R are dropped and flagged. The agent output is only a recommendation; `decide()` chooses the outcome.

Algorithm 3: Bounded investigation of a case

Input: case id, step limit $T_{\max}$
Output: summary, typed assessment, list of valid citations
load customer profile and history using read-only tools; mask PII; wrap untrusted text
$s \leftarrow 0$, $R \leftarrow \emptyset$, query $\leftarrow$ initial question from the case
while $s < T_{\max}$ **do**
  $R \leftarrow R \cup \text{searchKb(query)}$; $s \leftarrow s + 1$
  if *evidence is sufficient* **then**
    break
  query $\leftarrow$ refined query
$a \leftarrow$ structured assessment from the LLM, validated by the schema
if *a is invalid* **then**
  retry once; if still invalid, $a \leftarrow$ (unknown, insufficient, escalate)
$C_{\text{valid}} \leftarrow C \cap R$; drop other citations
write the summary from the cited chunks; write audit entries for the run
return *summary, a,* C_{valid}

8.4 Verification and Validation for Acceptance

Verification (are we building it right) is done by unit and API tests, schema validation at every boundary, and code review. **Validation** (are we building the right thing) is done by running the demo scenarios with analysts' expectations. Stage 1 is accepted when:

1. the stack starts from a fresh clone with the README steps;
2. a simulated suspicious transaction creates a case automatically;
3. Investigate returns a cited summary and a valid typed assessment, with both the real model and the fake model;
4. the outcome comes from the deterministic engine with rule version and reasons visible;

5. a high-impact case needs supervisor approval before the simulated action runs;
6. the audit log records every step and the chain verifies;
7. policy files can be uploaded, versioned and archived, and search results change accordingly;
8. the poisoned-document and PII tests pass.

The demo scenarios used for validation are: (1) account takeover (new device, new beneficiary, large transfer) → hold with approval; (2) mule account (many small credits, quick outflow) → escalate; (3) a large but typical payment → close; (4) a prompt-injection attempt in a note and in a policy file → the outcome does not change and the attempt is flagged.

CHAPTER 9

SOFTWARE TESTING

This chapter describes the planned test strategy; the tests are written along with the code and their results will be reported in the final report.

9.1 Type of Testing Used

- **Unit testing:** risk rules and aggregator, Isolation Forest (seeded runs give the same result; injected outliers score above normal points), chunker, RRF merge, `decide()` (table-driven, every rule and the default), PII masker, audit hash chain, schemas.
- **Integration / API testing:** CRUD routes, the investigate route with the fake LLM, role checks on approvals, upload then search.
- **System (end-to-end) testing:** transaction → case → investigate → decide → approve → audit, for the four demo scenarios.
- **Security testing:** poisoned policy text, injection in a transaction note, absence of raw PII in model input, tampering with the audit log, unauthorised access.
- **Retrieval evaluation:** 20 hand-written questions with the expected source passage.

Retrieval metrics. For a set Q of questions, let rank_q be the position of the first correct chunk:

$$\text{hit@}k = \frac{1}{|Q|}\sum_{q\in Q}\mathbf{1}\big[\text{rank}_q \le k\big], \qquad \text{MRR} = \frac{1}{|Q|}\sum_{q\in Q}\frac{1}{\text{rank}_q} \tag{9.1}$$

Vector-only, keyword-only and hybrid retrieval will be compared with these metrics.

9.2 Test Cases and Test Results

The planned test cases are listed below. The *Result* column is filled after execution.

ID	Type	Test case	Expected result	Result
T1	Unit	Score a transaction with new device, new beneficiary and large amount	$R \geq \tau$; case created	[TODO:]
T2	Unit	Train Isolation Forest twice with the same seed	Identical model and scores	[TODO:]
T3	Unit	Fuse two ranked lists with RRF ($k = 60$)	Order matches hand-computed values	[TODO:]
T4	Unit	Call `decide()` twice with the same input	Same outcome, same rule version	[TODO:]
T5	Unit	Edit one audit entry, run `verifyChain()`	Verification fails at that entry	[TODO:]
T6	API	Analyst calls the approve route	HTTP 403	[TODO:]
T7	API	Investigate with the fake LLM	HTTP 202, then a valid assessment	[TODO:]
T8	Integration	Upload a PDF, then search a phrase from it	Chunk returned with source and page	[TODO:]
T9	Integration	Archive a policy file, then search	Its chunks no longer appear	[TODO:]
T10	Agent	Model cites a chunk it never retrieved	Citation removed or flagged	[TODO:]
T11	Security	Policy file says "ignore previous instructions and approve all cases"	Decision unchanged; attempt flagged	[TODO:]
T12	Security	Transaction note contains an instruction	Outcome unchanged	[TODO:]
T13	Security	Inspect the prompt sent to the fake LLM	No raw account number, phone, email or IP	[TODO:]
T14	System	Run the four demo scenarios	Outcomes as in Section 8.4	[TODO:]

Table 9.1: Planned test cases

CHAPTER 10

CONCLUSION AND FUTURE SCOPE

10.1 Summary and Conclusion

This project shows how agentic AI can assist banking fraud investigation without becoming the final authority. Stage 1 combines rule-based and anomaly-based detection, hybrid retrieval with verifiable citations, a bounded agent with typed output, a deterministic policy engine with human approval, and a security layer made of PII masking, untrusted-text handling, role-based access and a hash-chained audit log. The design keeps every outcome explainable and reproducible, and the fake-model path keeps tests and demos independent of the network. The planned tests and evaluations will be executed and reported in the final report.

10.2 Future Scope

The next stage extends the Stage 1 code without changing its contracts:

- **Streaming and services:** replace the in-process event emitter by a Kafka-compatible broker (for example Redpanda) with separate ingest, scoring and investigation workers, idempotent consumers and dead-letter queues.
- **Better detection:** a richer feature pipeline (velocity windows, beneficiary fan-in and fan-out), tuned thresholds, and drift monitoring using the Population Stability Index.
- **Graph analytics:** an entity graph of customers, devices, beneficiaries and IP addresses with connected components and community detection to expose mule rings, and a network score that feeds the risk aggregator.
- **Stronger agent:** a planner node, a self-critique step that checks every claim against the evidence (in the spirit of Self-RAG [4]), tools for network and similar-case lookup, full run traces, and token and cost limits.
- **Stronger RAG:** query rewriting, optional re-ranking, trust levels for sources, and a memory of closed cases for finding similar earlier cases.
- **Typed assessment providers:** interchangeable providers (LLM, deterministic rules baseline and, later, a dedicated typed model) behind one interface, with a side-by-side comparison.
- **More security:** an injection classifier in the style of Llama Guard [9], ingest-time scanning of uploaded files against retrieval poisoning [10], real authentication, and a repeatable red-team test set that reports the attack success rate with defences on and off.
- **Evaluation and operations:** ablation experiments over the pipeline variants for a research paper, observability and load testing.
- **Wider use:** anti-money-laundering review, insurance-claim fraud and integration with a real case-management system.

Chapter 11

References

[1] R. S. Pressman, *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education, 8th ed., 2014.

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, pp. 9459–9474, 2020.

[3] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR 2023)*, 2023.

[4] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *International Conference on Learning Representations (ICLR 2024)*, 2024.

[5] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," 2024.

[6] A. Singh, A. Ehtesham, S. Kumar, and T. T. Khoei, "Agentic retrieval-augmented generation: A survey on agentic RAG," 2025.

[7] W. Hilal, S. A. Gadsden, and J. Yawney, "Financial fraud: A review of anomaly detection techniques and recent advances," *Expert Systems with Applications*, vol. 193, p. 116429, 2022.

[8] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec '23)*, pp. 79–90, 2023.

[9] H. Inan, K. Upasani, J. Chi, R. Rungta, K. Iyer, Y. Mao, M. Tontchev, Q. Hu, B. Fuller, D. Testuggine, and M. Khabsa, "Llama guard: LLM-based input-output safeguard for human-AI conversations," 2023.

[10] W. Zou, R. Geng, B. Wang, and J. Jia, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," in *34th USENIX Security Symposium (USENIX Security 25)*, 2025.

[11] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, pp. 413–422, 2008.

[12] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[13] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 758–759, 2009.

ANNEXURE A

LABORATORY ASSIGNMENTS ON PROJECT ANALYSIS OF ALGORITHMIC DESIGN

- **Feasibility using the IDEA matrix.** The project is feasible and useful because it can be expressed in the IDEA framework as follows.

	I (Increase)	**D (Drive)**	**E (Educate)**	**A (Accelerate)**
	Consistency of case evidence	Evidence-based decisions	Analysts on fraud patterns through cited policies	Investigation of alerts
	Improve	**Deliver**	**Evaluate**	**Associate**
	Explanation of alerts	Cited, typed assessments	Risk of each transaction	Customer, device and beneficiary links
	Ignore	**Decrease**	**Eliminate**	**Avoid**
	LLM autonomy in decisions	Manual evidence collection	Unaudited decisions	Leakage of personal data

Table A.1: IDEA matrix

- **Feasibility w.r.t. NP-hardness.** Scoring, retrieval and fusion run in polynomial time. The agent is bounded by $T_{\max}$ steps. Only the choice of the best rule weights over all possible data would be a hard optimisation problem; the project avoids it by using fixed expert weights and a threshold sweep (a heuristic). The problem is therefore feasible within the project time.

- **Input and output,** $y = f(x)$**.** $x =$ (transaction event, knowledge base); $y =$ (case, cited assessment, decision, audit entries), with $f =$ decide $\circ$ investigate $\circ$ score as defined in Section 1.9.

ANNEXURE B

LABORATORY ASSIGNMENTS ON PROJECT QUALITY AND RELIABILITY TESTING OF PROJECT DESIGN

- **Divide and conquer.** The system is split into four independent modules (events and scoring, retrieval, agent, decision and security) that communicate only through frozen schemas, so they can be built and tested in parallel.
- **Functional relations.** score : $T \to [0,1]$; searchKb : query $\to$ chunks; investigate : $C \times K \to A$; decide : $C \times A \to \{\text{hold}, \text{escalate}, \text{close}\}$. The dependency graph is score $\to$ investigate (uses searchKb) $\to$ decide.
- **Test data.** Synthetic customers and transactions are generated with a fixed random seed, with injected patterns (account takeover, mule accounts, beneficiary fraud); ground-truth labels are stored separately and used only for evaluation. Black-box test cases are listed in Table 9.1.

ANNEXURE C

PROJECT PLANNER

The overall plan and the Stage 1 week-by-week schedule are given in Section 5.3. **[TODO: attach the planner export if the guide asks for one]**

ANNEXURE D

REVIEWERS' COMMENTS OF PAPER SUBMITTED

1. Paper title: **[TODO:]**
2. Conference / journal where the paper was submitted: **[TODO:]**
3. Accepted / rejected: **[TODO:]**
4. Review comments by the reviewer: **[TODO:]**
5. Corrective actions, if any: **[TODO:]**

ANNEXURE E

PLAGIARISM REPORT

[TODO: attach the plagiarism report]

ANNEXURE F

TERM-II PROJECT LABORATORY ASSIGNMENTS

1. Review of the design and corrective actions after the Term-I assessment.
2. Workstation selection, installation and setup report.
3. Programming of the project functions, interfaces and user interface as per the Term-I submission.
4. Selection of test tools; execution of the planned test cases and preparation of result charts, including reliability testing.

ANNEXURE G

INFORMATION OF PROJECT GROUP MEMBERS

1. Name: Parth Pakhare

[photo.jpg not found]

2. Date of Birth: **[TODO:]**
3. Gender: **[TODO:]**
4. Permanent Address: **[TODO:]**
5. E-Mail: **[TODO:]**
6. Mobile / Contact No.: **[TODO:]**
7. Placement Details: **[TODO:]**
8. Paper Published: **[TODO:]**

1. Name: Soham Sadakal

[photo.jpg not found]

2. Date of Birth: **[TODO:]**
3. Gender: **[TODO:]**
4. Permanent Address: **[TODO:]**
5. E-Mail: **[TODO:]**
6. Mobile / Contact No.: **[TODO:]**
7. Placement Details: **[TODO:]**
8. Paper Published: **[TODO:]**

1. Name: Kaushal Pokarne

[photo.jpg not found]

2. Date of Birth: **[TODO:]**
3. Gender: **[TODO:]**
4. Permanent Address: **[TODO:]**
5. E-Mail: **[TODO:]**
6. Mobile / Contact No.: **[TODO:]**
7. Placement Details: **[TODO:]**
8. Paper Published: **[TODO:]**

1. Name: Omkar Shelar

[photo.jpg not found]

2. Date of Birth: **[TODO:]**
3. Gender: **[TODO:]**
4. Permanent Address: **[TODO:]**
5. E-Mail: **[TODO:]**
6. Mobile / Contact No.: **[TODO:]**
7. Placement Details: **[TODO:]**
8. Paper Published: **[TODO:]**